

VentureGraph — SSRN Paper

Abstract

VentureGraph 2.0 · The FinTech Foundry Edition

C-DE422 · Big Data Engineering II · Egypt University of Informatics
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Smart Money Silence: An Inverse Link-Prediction Signal for Private-to-Public Alpha Bridges

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Abstract (250 words)

I introduce **Smart Money Silence (SMS)**, an inverse link-prediction signal that operationalizes the structural *absence* of expected co-investment edges as a quantifiable feature of a directed, time-weighted venture-capital graph. SMS inverts the standard Lü–Zhou (2011) link-prediction paradigm: instead of scoring edges that are *likely to form*, SMS scores edges *that were expected to form but did not*. Formally, I identify

the top-tier investors in a Series A round (by pre-registered Temporal Precursor Score, TPS, at the eval date) and measure their absence from the matched Series B round. I propose the sector-aggregated silence rate as a candidate bearish signal for public-equity sector returns.

Using a Crunchbase 2013 snapshot (12,419 investments, 4,035 investors, 5 sectors), I compute SMS scores at quarterly resolution 2006–2013 and test the pre-registered H3 hypothesis that elevated sector SMS predicts negative sector alpha (Fama-French 5-factor residual) at horizons $k \in \{6, 12, 18, 24, 36\}$ months. **The empirical test fails to reject the null** at every horizon (Pearson $r \in [-0.06, +0.05]$, $p \in [0.61, 0.90]$, $n \in [35, 76]$). Post-hoc power analysis indicates the observed effect size ($|r| \approx 0.06$) requires $n \approx 2 \times 10^3$ to detect at $\alpha = 0.05$, 80% power — the present study is under-powered by $\sim 30\times$.

The contribution is therefore **methodological**: I provide the first operational specification of inverse link-prediction as a financial signal, embedded in a SHA-256-hash-locked pre-registered protocol with expanding-window contamination control and two-way clustered panel regression with wild-cluster bootstrap. The framework is independently valuable wherever graph topology, time, and finance intersect.

Keywords: link prediction, venture capital networks, Fama-French alpha, pre-registration, graph signal processing, point-in-time computation

JEL classification: G11, G14, G24, C46, C81

Extended summary for reviewers (1 page)

The problem

Classical link prediction (Lü & Zhou 2011; Liben-Nowell & Kleinberg 2007; Adamic & Adar 2003) answers the question: *given the current graph, which missing edges are most likely to appear next?* The entire literature treats edge *formation* as the signal. But in financial networks, the dog that doesn't bark often carries more information than the one that does. When prescient capital that *should* follow a promising Series A *declines to participate* in the Series B, that absence is structural — and potentially predictive of the underlying asset's trajectory.

The construction

I formalize three objects on a directed, time-weighted VC co-investment graph $G_t = (V_t, E_t, w_t)$:

Temporal Precursor Score (TPS). Volume-independent prescience score: $TPS(A) = |\text{portfolio}(A)|^{-1} \sum_c \sum_{B \in \text{top-tier}} w(A \rightarrow B, c) \cdot \text{rank}^{-1}(B)$. Captures who consistently enters *before* top-tier follow-on capital.

Swarm Signal Intensity (SSI). Rolling aggregate of TPS-weighted new entrants into a sector, measuring prescient capital inflow.

Smart Money Silence (SMS). The novel contribution: for each sector-quarter, the rate at which top-tier (top-30% TPS at the eval date) Series A investors *decline* to participate in matched Series B rounds.

The audit backbone

Every score is computed point-in-time on the expanding-window graph state and emits a SHA-256 hash tuple (`input_hash`, `protocol_hash`, `output_hash`, `eval_date`). The full protocol (hyperparameters, hypotheses, robustness tests) is pre-registered as a Python module, hashed, and git-committed before any empirical test. This is rare even in the published applied-econometrics literature and absent from every commercial VC-data platform I have examined.

The (null) empirical finding

Across five horizons and five sector ETFs, SMS does not predict FF5 abnormal returns at conventional significance. Correlation signs reverse between 12/24-month horizons (weakly consistent with the hypothesis) and 6/18/36-month horizons (counter-directional). Power analysis shows the sample is too small by a factor of $\sim 30\times$ for the observed effect size.

Why the null matters

The honest reporting of this null is part of the contribution. The framework — *inverse link-prediction as an audit-grade, point-in-time, pre-registered financial signal* — is deployable on any larger dataset (WRDS VentureXpert, Pitchbook, Magnitt MENA), and the first study with $n \geq 500$ per horizon will provide decisive evidence one way or the other. The present work proposes the construct, establishes the evaluation machinery, and documents that the construct is cleanly measurable.

Why it is not a commercial product claim

No part of this paper asserts the signal *works* in the sense of producing tradeable alpha in the 2013 Crunchbase sample. Claims to that effect would be unsupported. The companion dashboard's **Sovereign Mode** (`sovereign_mode.py`) demonstrates the governance and audit features of the framework — which are independently valuable to a sovereign wealth fund's compliance officer — but the dashboard does not and cannot assert alpha in this sample. The forward-looking extension to a larger dataset is labeled explicitly as future work.

What's next

Three experiments rank-order the future work:

1. Re-run on WRDS VentureXpert 2000–2024 for $n \geq 500$ per horizon.
2. Re-specify SMS at the deal level (each Series B as one observation) for $N \sim 10^4$.
3. Weight silence by the *individual silent investor's* TPS, not a uniform top-tier weight.

Any one of these tests can decisively resolve the empirical question.

Pre-registration hash (to be inserted after running `python preregistration.py`):

```
Protocol Hash : [RUN `python preregistration.py` TO GENERATE] Sealed At :  
[AUTO-FILLED ON RUN] Git Commit : [RECORD MANUALLY AFTER COMMIT]
```

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